

5 Project Development and Advancement

This section solves the ex ante advancement decision at a fixed conjectured qualified manufacturing-service price p_m . It uses the deterministic optimized route value defined above. CMO market consistency and the feedback that selects p_m^* are imposed in the capacity-market section below.

5.1 Innovation-object boundary and planned-project technology

Developer i chooses one common control $x_i \geq 0$. Its canonical meaning is project-development / advancement intensity: it advances viable drug projects to a commercialization-relevant route-planning stage. It excludes patent applications, patent-generating effort, basic research, and up-stream scientific discovery. There is no class-specific control x_{ig} .

Predetermined project-advancement productivity $a_i > 0$ converts the control into the expected measure of viable projects reaching route planning:

$$\lambda_i^{\text{plan}} = a_i x_i. \quad (1)$$

The control has units X, and a_i has units P/X, so λ_i^{plan} has units P.

Let $\kappa > 0$ be the cost scale and $\nu > 0$ its curvature parameter. The advancement-cost function is

$$C_X(x_i) = \frac{\kappa}{1+\nu} x_i^{1+\nu}, \quad \kappa > 0, \quad \nu > 0. \quad (2)$$

Here κ has units C/X^{1+ν}, so C_X is measured in currency per developer and decision cohort. Its marginal cost is $C'_X(x_i) = \kappa x_i^\nu$. The boundary $\nu = 1$ gives a quadratic advancement cost.

5.2 Expected optimized project value

For developer i , institutional regime M , and fixed conjectured price p_m , define

$$\begin{aligned} \Omega_i(M, p_m) &= \mathbb{E}_F[W_i(q, m; M, p_m)] \\ &= \int W_i(q, m; M, p_m) dF(q, m). \end{aligned} \quad (3)$$

The subscript i includes the dependence of route value on developer characteristics, especially manufacturing capability k_i . Because the route-value maximum contains abandonment with value zero, $W_i(q, m; M, p_m) \geq 0$. Assume measurable route values and finite expected optimized value. A sufficient condition is integrability of the sum of nonnegative gross operating values of feasible routes and the positive part of T . Then

$$0 \leq \Omega_i(M, p_m) < +\infty.$$

The expectation averages the optimized deterministic project value.

5.3 Ex ante advancement problem

The developer solves

$$\max_{x_i \geq 0} \left\{ \beta a_i x_i \Omega_i(M, p_m) - \frac{\kappa}{1+\nu} x_i^{1+\nu} \right\}. \quad (4)$$

The decision precedes the (q, m) draw and downstream route choice. Conditional on M, p_m, a_i, k_i , the expected value $\Omega_i(M, p_m)$ is therefore fixed with respect to the individual choice x_i .

The necessary and sufficient Kuhn–Tucker conditions are

$$\begin{aligned} x_i &\geq 0, & \kappa x_i^\nu - \beta a_i \Omega_i(M, p_m) &\geq 0, \\ x_i [\kappa x_i^\nu - \beta a_i \Omega_i(M, p_m)] &= 0. \end{aligned} \quad (5)$$

For an interior solution, the FOC is

$$\kappa x_i^\nu = \beta a_i \Omega_i(M, p_m). \quad (6)$$

For $x_i > 0$, the objective's second derivative is

$$-\kappa \nu x_i^{\nu-1} < 0. \quad (7)$$

More generally, $x_i^{1+\nu}$ is strictly convex on the nonnegative domain for every $\nu > 0$. Hence the objective is strictly concave. Its superlinear cost dominates the linear benefit as $x_i \rightarrow \infty$, so a unique maximizer exists.

The unique optimal common advancement intensity is

$$x_i^*(M, p_m) = \left[\frac{\beta a_i}{\kappa} \Omega_i(M, p_m) \right]^{1/\nu}. \quad (8)$$

If $\Omega_i = 0$, the formula gives the unique corner $x_i^* = 0$. If $\Omega_i > 0$, the solution is interior. Dimensionally, $\beta a_i \Omega_i / \kappa$ has units X^ν , so the solution has units X .

5.4 Fixed-price binary MAH channel

At a fixed p_m , deterministic route choice implies the pointwise weak ordering

$$W_i(q, m; 1, p_m) \geq W_i(q, m; 0, p_m).$$

Taking expectations gives the finite policy comparison

$$\Omega_i(1, p_m) - \Omega_i(0, p_m) = \int [W_i(q, m; 1, p_m) - W_i(q, m; 0, p_m)] dF(q, m) \geq 0. \quad (9)$$

For positive Ω_i ,

$$\frac{\partial x_i^*}{\partial \Omega_i} = \frac{1}{\nu} \left(\frac{\beta a_i}{\kappa} \right)^{1/\nu} \Omega_i^{1/\nu-1} > 0.$$

Thus, at fixed p_m ,

$$\Omega_i(1, p_m) > \Omega_i(0, p_m) \iff x_i^*(1, p_m) > x_i^*(0, p_m).$$

The strict inequality in expected value requires a positive-probability set of projects for which the newly feasible entrusted route strictly improves the pre-reform maximum. Otherwise both expected value and advancement can have a zero reform effect.

This is a binary finite comparison, not a derivative with respect to M . It is also a fixed-price result, not the equilibrium-price comparative static through p_m^* . The regime changes neither a_i , κ , ν , the cost function, nor the project distribution. MAH affects x_i^* only through Ω_i . Manufacturing capability k_i affects Ω_i through route values and sorting, not through $a_i x_i$ or $C_X(x_i)$.

5.5 Downstream project-value gap

For an appendix-only value decomposition, let B_i denote the continuation-value benchmark before one additional viable project reaches route planning, and let K_i denote value after it reaches that stage. Define

$$K_i = B_i + \Omega_i, \quad K_i - B_i = \Omega_i. \quad (10)$$

Substituting this accounting identity into the interior FOC gives

$$C'_X(x_i) = \beta a_i(K_i - B_i). \quad (11)$$

This is an Akcigit-style downstream project-value interpretation. The auxiliaries B_i and K_i are not recursive state variables, and the identity does not claim that MAH directly raises scientific-research productivity.

5.6 Scope boundary

The baseline contains one common x_i , one cost function C_X , and one expected optimized project value Ω_i . It contains no separate research and development controls, no binding financial constraint, no patent outcome, no novelty-class-specific control, and no CMO market-clearing condition. Project advancement is ex ante; route choice and observed holder–producer separation are downstream.